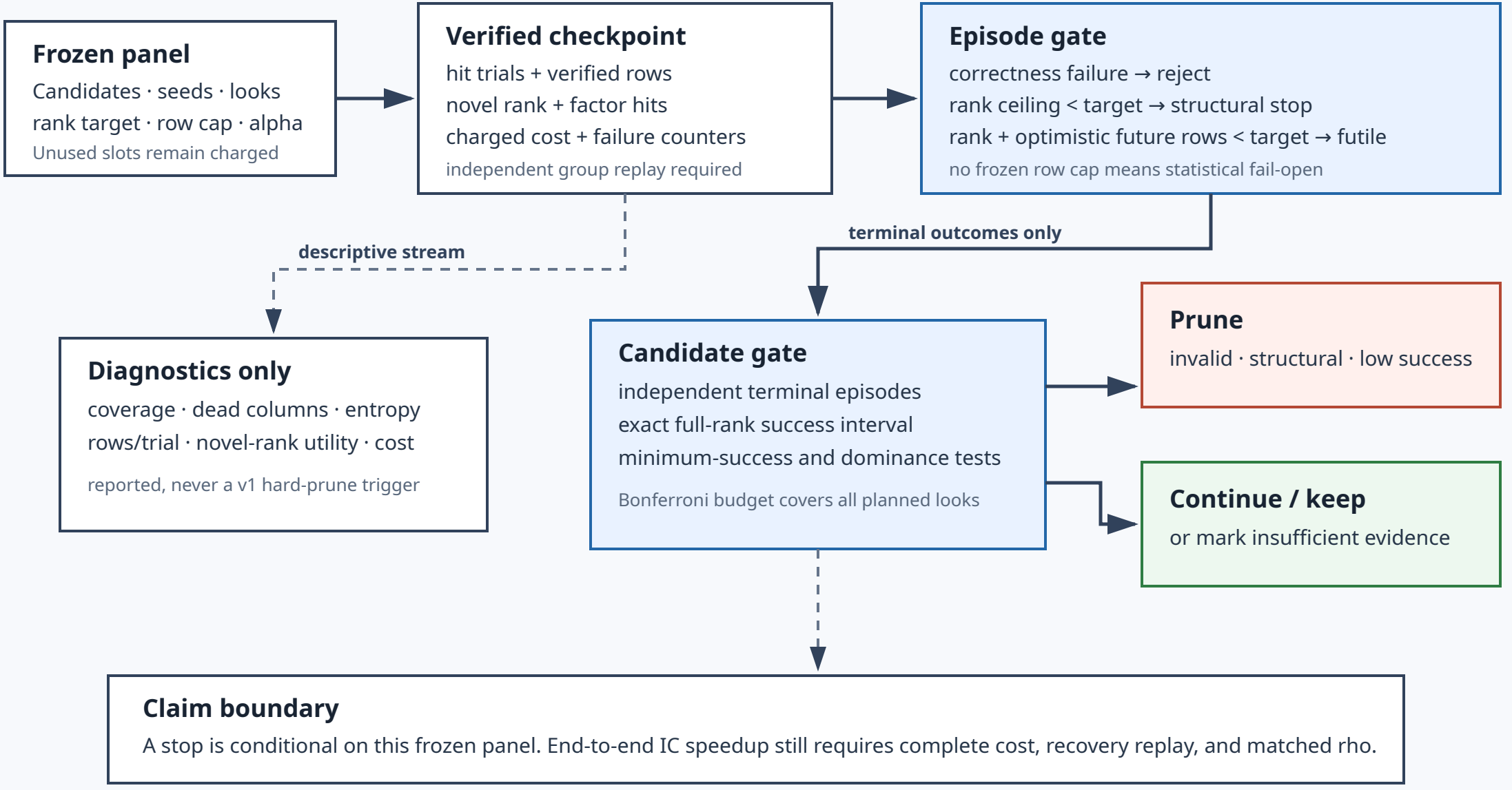Empirical pruning without promoting stage yield to a speedup